

Study Report



Analyzed File	Z - axis v8
Version	Autodesk Fusion (2702.1.58)
Creation Date	2026-05-06, 23:39:39
Author	mohan

☐ **Report Properties**

Title	Studies
Author	mohan

Simulation Model 1

Study 2 - Modal Frequencies

Study Properties

Study Type	Modal Frequencies
Last Modification Date	2026-05-06, 23:13:26

Settings

General

Contact Tolerance	0.10 mm
Number of Modes	8
Frequency Range [Hz]	20000-30000
Compute Preloaded Modes	No

Mesh

Average Element Size (% of model size)	
Solids	8
Scale Mesh Size Per Part	No
Average Element Size (absolute value)	-
Element Order	Parabolic
Create Curved Mesh Elements	No
Max. Turn Angle on Curves (Deg.)	60
Max. Adjacent Mesh Size Ratio	1.5
Max. Aspect Ratio	10
Minimum Element Size (% of average size)	20

Adaptive Mesh Refinement

Number of Refinement Steps	0
Results Convergence Tolerance (%)	20
Frequency Mode	1

Materials

Component	Material	Safety Factor
Frame 1 v1:1	Steel	Yield Strength
rail1 Copy v1:1	Steel	Yield Strength
Frame 2 v1:1	Steel	Yield Strength
rod v1:1	Steel	Yield Strength
rod v1:2	Steel	Yield Strength
motor v1:1/base:1	Steel	Yield Strength
motor v1:1/middle:1	Steel	Yield Strength
motor v1:1/top:1	Steel	Yield Strength
motor v1:1/rod:1	Steel	Yield Strength
Frame 1.2 v1:1	Steel	Yield Strength
Z- thread v1:1	Steel	Yield Strength
coupler v1:1	Steel	Yield Strength
bolt v1:1	Steel	Yield Strength
bolt v1:2	Steel	Yield Strength
bolt v1:3	Steel	Yield Strength
bolt v1:4	Steel	Yield Strength

bolt v1:5	Steel	Yield Strength
bolt v1:6	Steel	Yield Strength
bolt v1:7	Steel	Yield Strength
bolt v1:8	Steel	Yield Strength
Z- traveller v1:1	Steel	Yield Strength
trans and drill v1:1/connect v1:1	Steel	Yield Strength
trans and drill v1:1/connect v1:2	Steel	Yield Strength
trans and drill v1:1/horn v1:1	Steel	Yield Strength
trans and drill v1:1/Transducer v1:1/Body1	Steel	Yield Strength
trans and drill v1:1/Transducer v1:1/Body2	Steel	Yield Strength
trans and drill v1:1/Transducer v1:1/Body3	Steel	Yield Strength
trans and drill v1:1/Transducer v1:1/Body4	Steel	Yield Strength
trans and drill v1:1/Transducer v1:1/Body5	Nickel-Copper Alloy 400	Yield Strength
trans and drill v1:1/Transducer v1:1/Body6	Nickel-Copper Alloy 400	Yield Strength
trans and drill v1:1/drill-1 v3 v1:1	Titanium 6Al-4V	Yield Strength
trans and drill v1:1/spindle rotory v1:1/bearing v1:1	Steel	Yield Strength
trans and drill v1:1/spindle rotory v1:1/main v1:1	Steel	Yield Strength
trans and drill v1:1/spindle rotory v1:1/p-1(connect) v1:1	Steel	Yield Strength
Component2:1	Steel	Yield Strength
Flange P v1:1	Steel	Yield Strength
Flange P v1:2	Steel	Yield Strength

☐ **Nickel-Copper Alloy 400**

Density	8.830E-06 kg / mm ³
Young's Modulus	179300.00 MPa
Poisson's Ratio	0.315
Yield Strength	220.00 MPa
Ultimate Tensile Strength	558.00 MPa
Thermal Conductivity	0.401 W / (mm C)
Thermal Expansion Coefficient	1.090E-05 / C
Specific Heat	502.00 J / (kg C)

☐ **Steel**

Density	7.850E-06 kg / mm ³
Young's Modulus	210000.00 MPa
Poisson's Ratio	0.30
Yield Strength	207.00 MPa
Ultimate Tensile Strength	345.00 MPa
Thermal Conductivity	0.056 W / (mm C)
Thermal Expansion Coefficient	1.200E-05 / C
Specific Heat	480.00 J / (kg C)

☐ **Titanium 6Al-4V**

Density	4.430E-06 kg / mm ³
Young's Modulus	113763.35 MPa
Poisson's Ratio	0.35
Yield Strength	882.528 MPa
Ultimate Tensile Strength	1034.213 MPa
Thermal Conductivity	0.007 W / (mm C)
Thermal Expansion Coefficient	8.600E-06 / C
Specific Heat	526.30 J / (kg C)

☐ Contacts

☐ Bonded

Name
[S] Bonded1 [p-1(connect) v1:1 Flange P v1:1]
[S] Bonded2 [p-1(connect) v1:1 Flange P v1:1]
[S] Bonded3 [p-1(connect) v1:1 Flange P v1:1]
[S] Bonded4 [p-1(connect) v1:1 Flange P v1:1]
[S] Bonded5 [p-1(connect) v1:1 Flange P v1:1]
[S] Bonded6 [p-1(connect) v1:1 Flange P v1:2]
[S] Bonded7 [p-1(connect) v1:1 Flange P v1:2]
[S] Bonded8 [p-1(connect) v1:1 Flange P v1:2]
[S] Bonded9 [p-1(connect) v1:1 Flange P v1:2]
[S] Bonded10 [p-1(connect) v1:1 Flange P v1:2]
[S] Bonded11 [bearing v1:1 main v1:1]
[S] Bonded12 [bearing v1:1 p-1(connect) v1:1]
[S] Bonded13 [bearing v1:1 p-1(connect) v1:1]
[S] Bonded14 [bearing v1:1 p-1(connect) v1:1]
[S] Bonded15 [Transducer v1:1(Body3) Transducer v1:1(Body6)]
[S] Bonded16 [Transducer v1:1(Body2) Transducer v1:1(Body5)]
[S] Bonded17 [Transducer v1:1(Body2) Transducer v1:1(Body3)]
[S] Bonded18 [Transducer v1:1(Body2) Transducer v1:1(Body4)]
[S] Bonded19 [Transducer v1:1(Body1) Transducer v1:1(Body3)]
[S] Bonded20 [horn v1:1 drill-1 v3 v1:1]
[S] Bonded21 [horn v1:1 Transducer v1:1(Body1)]
[S] Bonded22 [connect v1:2 Transducer v1:1(Body1)]
[S] Bonded23 [connect v1:2 Transducer v1:1(Body4)]
[S] Bonded24 [connect v1:2 Transducer v1:1(Body3)]
[S] Bonded25 [connect v1:2 Transducer v1:1(Body2)]
[S] Bonded26 [connect v1:2 Transducer v1:1(Body1)]
[S] Bonded27 [connect v1:2 Transducer v1:1(Body4)]
[S] Bonded28 [connect v1:2 Transducer v1:1(Body3)]
[S] Bonded29 [connect v1:2 Transducer v1:1(Body2)]
[S] Bonded30 [connect v1:2 Transducer v1:1(Body1)]
[S] Bonded31 [connect v1:1 connect v1:2]
[S] Bonded32 [connect v1:1 Transducer v1:1(Body1)]
[S] Bonded33 [connect v1:1 Transducer v1:1(Body1)]
[S] Bonded34 [Z- traveller v1:1 Flange P v1:1]
[S] Bonded35 [Z- traveller v1:1 p-1(connect) v1:1]
[S] Bonded36 [Z- traveller v1:1 Flange P v1:2]
[S] Bonded37 [Z- thread v1:1 Z- traveller v1:1]
[S] Bonded38 [Frame 1.2 v1:1 Z- traveller v1:1]
[S] Bonded39 [Frame 1.2 v1:1 Z- thread v1:1]
[S] Bonded40 [top:1 Frame 1.2 v1:1]
[S] Bonded41 [top:1 rod:1]
[S] Bonded42 [middle:1 top:1]
[S] Bonded43 [middle:1 top:1]
[S] Bonded44 [middle:1 top:1]
[S] Bonded45 [middle:1 top:1]
[S] Bonded46 [middle:1 rod:1]
[S] Bonded47 [middle:1 top:1]
[S] Bonded48 [middle:1 top:1]
[S] Bonded49 [middle:1 top:1]

[S] Bonded50 [middle:1 top:1]
[S] Bonded51 [middle:1 top:1]
[S] Bonded52 [middle:1 top:1]
[S] Bonded53 [middle:1 top:1]
[S] Bonded54 [middle:1 top:1]
[S] Bonded55 [middle:1 top:1]
[S] Bonded56 [base:1 middle:1]
[S] Bonded57 [base:1 middle:1]
[S] Bonded58 [base:1 middle:1]
[S] Bonded59 [base:1 middle:1]
[S] Bonded60 [base:1 middle:1]
[S] Bonded61 [base:1 middle:1]
[S] Bonded62 [base:1 middle:1]
[S] Bonded63 [base:1 middle:1]
[S] Bonded64 [base:1 middle:1]
[S] Bonded65 [base:1 middle:1]
[S] Bonded66 [base:1 middle:1]
[S] Bonded67 [base:1 middle:1]
[S] Bonded68 [base:1 middle:1]
[S] Bonded69 [rod v1:2 Component2:1]
[S] Bonded70 [rod v1:2 Z- traveller v1:1]
[S] Bonded71 [rod v1:2 Frame 1.2 v1:1]
[S] Bonded72 [rod v1:1 Component2:1]
[S] Bonded73 [rod v1:1 Z- traveller v1:1]
[S] Bonded74 [rod v1:1 Frame 1.2 v1:1]
[S] Bonded75 [Frame 2 v1:1 Frame 1.2 v1:1]
[S] Bonded76 [Frame 2 v1:1 top:1]
[S] Bonded77 [Frame 2 v1:1 bolt v1:5]
[S] Bonded78 [Frame 2 v1:1 bolt v1:7]
[S] Bonded79 [Frame 2 v1:1 bolt v1:6]
[S] Bonded80 [Frame 2 v1:1 bolt v1:8]
[S] Bonded81 [Frame 2 v1:1 top:1]
[S] Bonded82 [Frame 2 v1:1 rod v1:1]
[S] Bonded83 [Frame 2 v1:1 rod v1:1]
[S] Bonded84 [Frame 2 v1:1 rod v1:2]
[S] Bonded85 [Frame 2 v1:1 rod v1:2]
[S] Bonded86 [Frame 2 v1:1 bolt v1:5]
[S] Bonded87 [Frame 2 v1:1 bolt v1:5]
[S] Bonded88 [Frame 2 v1:1 bolt v1:7]
[S] Bonded89 [Frame 2 v1:1 bolt v1:7]
[S] Bonded90 [Frame 2 v1:1 bolt v1:6]
[S] Bonded91 [Frame 2 v1:1 bolt v1:6]
[S] Bonded92 [Frame 2 v1:1 bolt v1:8]
[S] Bonded93 [Frame 2 v1:1 bolt v1:8]
[S] Bonded94 [rail1 Copy v1:1 Frame 1.2 v1:1]
[S] Bonded95 [rail1 Copy v1:1 Frame 1.2 v1:1]
[S] Bonded96 [rail1 Copy v1:1 Frame 1.2 v1:1]
[S] Bonded97 [rail1 Copy v1:1 Frame 1.2 v1:1]
[S] Bonded98 [rail1 Copy v1:1 Frame 1.2 v1:1]
[S] Bonded99 [rail1 Copy v1:1 bolt v1:7]
[S] Bonded100 [rail1 Copy v1:1 bolt v1:1]
[S] Bonded101 [rail1 Copy v1:1 bolt v1:5]
[S] Bonded102 [rail1 Copy v1:1 bolt v1:4]

[S] Bonded103 [rail1 Copy v1:1 bolt v1:8]
[S] Bonded104 [rail1 Copy v1:1 bolt v1:2]
[S] Bonded105 [rail1 Copy v1:1 bolt v1:6]
[S] Bonded106 [rail1 Copy v1:1 bolt v1:3]
[S] Bonded107 [rail1 Copy v1:1 Frame 2 v1:1]
[S] Bonded108 [Frame 1 v1:1 Component2:1]
[S] Bonded109 [Frame 1 v1:1 rail1 Copy v1:1]
[S] Bonded110 [Frame 1 v1:1 bolt v1:1]
[S] Bonded111 [Frame 1 v1:1 bolt v1:4]
[S] Bonded112 [Frame 1 v1:1 bolt v1:2]
[S] Bonded113 [Frame 1 v1:1 bolt v1:3]
[S] Bonded114 [Frame 1 v1:1 rod v1:2]
[S] Bonded115 [Frame 1 v1:1 rod v1:2]
[S] Bonded116 [Frame 1 v1:1 rod v1:1]
[S] Bonded117 [Frame 1 v1:1 rod v1:1]
[S] Bonded118 [Frame 1 v1:1 bolt v1:1]
[S] Bonded119 [Frame 1 v1:1 bolt v1:1]
[S] Bonded120 [Frame 1 v1:1 bolt v1:4]
[S] Bonded121 [Frame 1 v1:1 bolt v1:4]
[S] Bonded122 [Frame 1 v1:1 bolt v1:2]
[S] Bonded123 [Frame 1 v1:1 bolt v1:2]
[S] Bonded124 [Frame 1 v1:1 bolt v1:3]
[S] Bonded125 [Frame 1 v1:1 bolt v1:3]

Mesh

Type	Nodes	Elements
Solids	1102811	710686

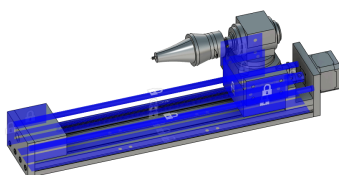
Load Case1

Constraints

Fixed1

Type	Fixed
Ux	Fixed
Uy	Fixed
Uz	Fixed

Selected Entities



Results

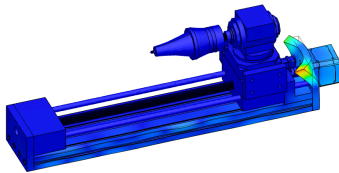
☐ Result Summary

Frequency	Participation X	Participation Y	Participation Z
Mode 1: 20125.051 Hz	0.465000002	0.212899991	0.82299998
Mode 2: 20604.57 Hz	0	0	0
Mode 3: 21108.09 Hz	0.0022	0.0052	0.0231
Mode 4: 21224.33 Hz	0.225399993	0.083899999	0.0159
Mode 5: 21664.99 Hz	0.254000002	3.26559991	1.45810004
Mode 6: 21906.619 Hz	0.0078	0.549500016	8.27400014
Mode 7: 22113.779 Hz	0.0003	0.286199991	0.42630001
Mode 8: 22248.73 Hz	0.0037	3.34560014	0.450500008

☐ Total Modal Displacement

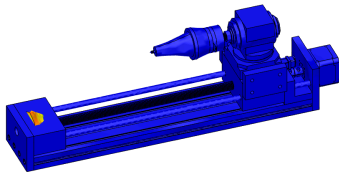
☐ Mode 1: 20125.051 Hz Total Modal Displacement

0.00  1.00



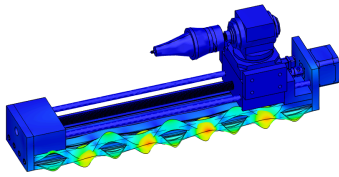
☐ Mode 2: 20604.57 Hz Total Modal Displacement

0.00  1.00



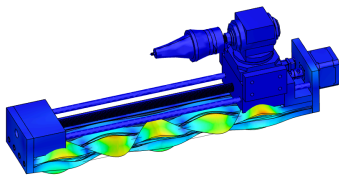
☐ Mode 3: 21108.09 Hz Total Modal Displacement

0.00  1.00



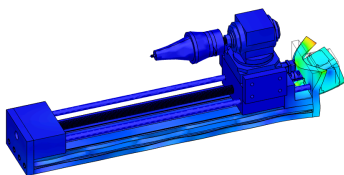
☐ **Mode 4: 21224.33 Hz Total Modal Displacement**

0.00  1.00



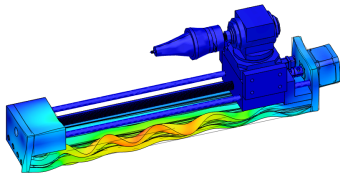
☐ **Mode 5: 21664.99 Hz Total Modal Displacement**

0.00  1.00



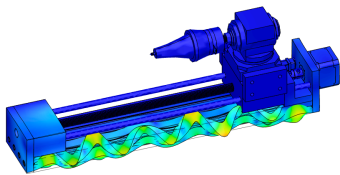
☐ **Mode 6: 21906.619 Hz Total Modal Displacement**

0.00  1.00



☐ **Mode 7: 22113.779 Hz Total Modal Displacement**

0.00  1.00



☐ **Mode 8: 22248.73 Hz Total Modal Displacement**

0.00  1.00

